# Thesis Proposal: Meteorological Determinants and Chemical Agent Dispersal Patterns: A Comprehensive Assessment of the March 1991 Khamisiyah Ammunition Storage Facility Demolition Events

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## Introduction

This thesis examines the atmospheric dispersal patterns of chemical warfare agents following the March 1991 demolition events at the Khamisiyah Ammunition Storage Facility in Iraq. Utilizing Geographic Information System (GIS) analysis, meteorological data, photographic evidence, and declassified military records, this research aims to reconstruct the actual dispersal patterns following both the March 4 Bunker 73 and March 10 "Pit" demolitions. The study will evaluate the validity of official modeling assumptions through comparative analysis with documented weather conditions, physical evidence, and established atmospheric dispersion principles. The findings will contribute to our understanding of chemical agent dispersal in complex weather systems and inform future methodological approaches to retrospective exposure assessment.

The demolition of chemical warfare agent-containing munitions at the Khamisiyah Ammunition Storage Facility represents a critical case study in post-conflict environmental exposure assessment. While official Department of Defense (DoD) and Central Intelligence Agency (CIA) modeling focused primarily on the March 10 "Pit" demolition, substantial evidence suggests that the March 4 Bunker 73 demolition may have resulted in more significant troop exposures due to differing meteorological conditions and troop positions. This research addresses the methodological discrepancies between official dispersal models and documented meteorological evidence through objective analysis of declassified records, meteorological data, and physical evidence.

Central to this research is the newly recovered complete meteorological dataset for March 4, 1991—data previously claimed to be nonexistent by the CIA and DoD Inspector General as a primary justification for not modeling the Bunker 73 demolition. This dataset represents a significant opportunity to comprehensively assess the atmospheric conditions present during the initial and potentially most consequential demolition

event at Khamisiyah. By applying contemporary atmospheric modeling techniques to this historical dataset, this research will provide new insights into the potential chemical agent exposure patterns that may have affected U.S. troops in the vicinity.

## Literature Review

### Chemical Warfare Agent Behavior and Transport Mechanisms

The environmental fate and transport of chemical warfare agents, particularly organophosphate nerve agents like sarin and cyclosarin found at Khamisiyah, has been extensively studied. Ellison (2008) demonstrated that the environmental behavior of nerve agents depends significantly on meteorological conditions, with temperature and humidity affecting degradation rates. Similarly, Talmage et al. (2007) found that sarin persistence is highly dependent on surface type, environmental conditions, and initial droplet size, with potential for longer-term environmental stability than previously assumed. These findings align with Kistner's (2017) conclusion that standard modeling approaches often underestimate the environmental persistence of nerve agents in complex field conditions.

Regarding transport mechanisms, Wayment et al. (2011) established that atmospheric transport of chemical agents is significantly affected by thermal stratification and turbulence patterns in boundary layers. This is particularly relevant to the Khamisiyah case, as noted by Tucker and Johnson (2018), who identified inconsistencies in historical plume modeling that failed to account for complex topographical and meteorological interactions. Hanna et al. (2009) demonstrated that mesoscale meteorological features, including frontal boundaries like those present in southern Iraq during early March 1991, can significantly modify predicted chemical agent dispersion patterns.

Research by Watson et al. (2015) on historical chemical weapons testing found that agent dispersion patterns often differ substantially from theoretical predictions due to previously unidentified meteorological phenomena. Similarly, Blewett (2018) identified systematic biases in retrospective exposure models that failed to incorporate comprehensive meteorological data. These findings are further supported by Mackay's (2020) analysis of historical chemical incidents, which revealed consistent underestimation of dispersion areas when models relied on simplified weather assumptions.

The gap in current knowledge centers on the application of these established principles to the specific conditions at Khamisiyah during March 1991. While the general behavior of chemical agents is well understood, no comprehensive study has applied contemporary modeling techniques to newly recovered meteorological data to reassess the March 4 Bunker 73 demolition event.

### Methodological Approaches to Retrospective Exposure Assessment

Retrospective exposure assessment methodologies have evolved significantly since the initial Khamisiyah investigations. Paustenbach (2012) established frameworks for comprehensive retrospective exposure

assessment that integrate multiple data sources including meteorological records, eyewitness accounts, and geographical information. Building on this foundation, Rhodes et al. (2016) developed specific methodological guidelines for chemical incident reconstruction that emphasize the importance of complete meteorological data at multiple elevations and time points.

Critical evaluations of previous exposure assessments have identified significant methodological limitations. The Government Accountability Office (2004) concluded that DoD's Khamisiyah exposure assessment methodology contained fundamental flaws, specifically noting that "the models were not fully developed for analyzing long-range dispersion of chemical warfare agents" and that source term data was "uncertain, incomplete, and nonvalidated." These criticisms align with Zhang and Peterson's (2018) systematic review of historical exposure assessments, which found that selective utilization of available data represents the most common source of error in retrospective analyses.

Brimblecombe and Makuch (2014) emphasized the importance of transparent methodological approaches in retrospective exposure assessment, highlighting that selection bias in data utilization can significantly alter conclusions. This concern is echoed by Cullen (2010), who established methodological standards for chemical exposure reconstruction that emphasize comprehensive data incorporation and explicit acknowledgment of uncertainty.

Recent methodological advances offer opportunities for improved assessment. Martinez et al. (2022) demonstrated the application of machine learning techniques to historical meteorological data for improved identification of complex weather patterns affecting chemical agent dispersion. Similarly, Johnson and Wright (2019) established protocols for integrating geographical information systems with atmospheric dispersion models to better account for terrain effects and troop positioning.

The methodological gap addressed by this research involves applying these contemporary standards and techniques to reassess the Khamisiyah incident with previously unavailable data. No study has yet comprehensively reanalyzed the March 4 Bunker 73 demolition using the complete meteorological dataset while applying current methodological standards for retrospective exposure assessment.

## Gulf War Illness and Chemical Exposure Correlations

The relationship between potential chemical agent exposure at Khamisiyah and subsequent health outcomes among veterans has been extensively studied. White et al. (2016) found that veterans potentially exposed to low-level chemical agents at Khamisiyah reported higher rates of chronic multisymptom illness compared to non-exposed veterans. These findings are supported by Sullivan et al. (2018), who identified neurological abnormalities among veterans previously considered exposed based on the revised 2000 Khamisiyah plume model.

Epidemiological studies by Steele et al. (2012) demonstrated that Gulf War illness rates were highest among veterans who served in areas with potential chemical exposures, with a dose-response relationship

observed relative to exposure duration. More recent research by Steele et al. (2024) confirmed that even 30+ years after the Gulf War, exposure to chemical warfare agents remains one of the strongest predictors of Gulf War illness, with 31% of veterans continuing to meet CDC criteria for severe symptoms.

Research on exposure classification methodology has revealed significant limitations. Bullman et al. (2005) found that veterans classified as exposed to Khamisiyah chemical agents had nearly twice the risk of brain cancer deaths compared to unexposed veterans, with risk increasing in proportion to exposure duration. However, Chao et al. (2010) demonstrated that exposure classification methods significantly affected study outcomes, with the 2000 plume model reclassification substantially changing which veterans were considered exposed.

Critical analysis by the Research Advisory Committee on Gulf War Veterans' Illnesses (2014) concluded that epidemiological studies were hampered by uncertainty regarding which troops were actually exposed, noting that "conclusions regarding health outcomes associated with Khamisiyah exposures are limited by questions concerning the accuracy of exposure models." Similarly, Haley et al. (2022) found that misclassification of exposure status based on incomplete modeling represents a significant limitation in understanding Gulf War illness etiology.

The gap addressed by this research involves improving exposure classification accuracy through more comprehensive modeling. By developing more accurate chemical agent dispersion models based on complete meteorological data for the March 4 Bunker 73 demolition, this study will provide a foundation for improved epidemiological research on the health effects associated with potential chemical exposures.

In summary, while significant research exists on chemical agent behavior, retrospective exposure assessment methodologies, and Gulf War illness correlations, no comprehensive study has applied contemporary modeling techniques to newly recovered meteorological data to reassess chemical agent dispersal patterns from the March 4, 1991 Bunker 73 demolition at Khamisiyah. This research addresses this gap by integrating multiple data sources with advanced atmospheric dispersion modeling to provide a more accurate understanding of potential troop exposures.

## Research Questions/Aims/Hypotheses

1. **Research Question 1:** What meteorological conditions were present at Khamisiyah on March 4, 1991, and how did they differ from conditions on March 10, 1991? **Aim:** To develop a comprehensive meteorological profile for both demolition events using newly recovered data and evaluate the validity of weather assumptions used in official modeling. **Hypothesis:** The newly recovered meteorological data will demonstrate significant discrepancies from the assumptions used in official modeling, particularly regarding wind direction, atmospheric stability, and precipitation timing on March 4, 1991. This is relevant because accurate meteorological data is fundamental to understanding chemical agent dispersion patterns and will establish the foundation for reassessing potential troop exposures.

2. **Research Question 2:** How do the documented meteorological conditions affect chemical agent dispersal patterns when integrated into contemporary atmospheric dispersion models? **Aim:** To develop revised chemical agent dispersion models for both the March 4 Bunker 73 and March 10 Pit demolitions using multiple contemporary modeling approaches. **Hypothesis:** Revised modeling based on complete meteorological data will demonstrate significantly different chemical agent dispersion patterns than official assessments, particularly for the March 4 Bunker 73 demolition. This is relevant because accurate dispersion modeling is essential for determining which troops may have been exposed to chemical agents and at what concentrations, potentially affecting healthcare eligibility for thousands of veterans.
3. **Research Question 3:** How does the integration of troop location data with revised dispersion models affect estimates of potential chemical agent exposure among U.S. forces? **Aim:** To conduct geospatial analysis of documented troop positions relative to modeled chemical agent dispersion patterns for both demolition events. **Hypothesis:** Integration of revised dispersion models with comprehensive troop location data will identify significantly more personnel potentially exposed to chemical agents than officially acknowledged, particularly from the March 4 Bunker 73 demolition. This is relevant because accurate exposure assessment is critical for epidemiological research on Gulf War illness and for ensuring appropriate healthcare and compensation for affected veterans.

## Intellectual Merit and Broader Impact

### Intellectual Merit

This research will advance scientific knowledge in multiple disciplines through innovative integration of meteorological analysis, atmospheric dispersion modeling, and geospatial techniques applied to historical data. By applying contemporary methodological standards to newly recovered meteorological data, this study represents the first comprehensive scientific reassessment of the March 4, 1991 Bunker 73 demolition at Khamisiyah using complete weather records previously claimed to be nonexistent.

The project will contribute significantly to the atmospheric sciences by demonstrating methodological approaches for reconstructing historical chemical agent dispersion patterns in complex weather systems. It will establish protocols for integrating multiple atmospheric modeling techniques with limited historical data to provide more robust and comprehensive assessments than single-model approaches. The application of machine learning algorithms to identify pattern relationships in historical meteorological data represents a novel approach to understanding complex weather dynamics in retrospective analyses.

In the field of retrospective exposure assessment, this research will advance methodological standards by developing frameworks for addressing instances of selective data utilization in historical analyses. By conducting side-by-side comparative analysis of multiple modeling approaches, the study will establish best practices for quantifying and communicating uncertainty in historical exposure reconstructions. This

methodological contribution extends beyond the specific Khamisiyah case to inform approaches for assessing other historical hazardous material releases.

For GIS and spatial analysis, the research will demonstrate innovative techniques for integrating historical troop movement data with atmospheric dispersion models. This includes developing methodologies for addressing spatiotemporal uncertainty in historical positioning data and establishing protocols for quantifying exposure likelihood based on relative positioning within modeled plume areas. These approaches will advance spatial analysis methodologies for historical exposure assessment more broadly.

The interdisciplinary nature of this research—integrating meteorology, toxicology, GIS, and historical analysis—represents an innovative approach to addressing complex historical exposure events. By establishing clear methodological frameworks for integrating these diverse disciplines, the project will advance scientific approaches to similar complex retrospective assessments.

## Broader Impact

This research has significant potential to benefit society, particularly the hundreds of thousands of Gulf War veterans potentially affected by chemical agent exposures at Khamisiyah. By providing a more accurate and comprehensive assessment of potential exposure patterns, this research will directly inform healthcare and compensation decisions affecting these veterans. Recent research confirms that 31% of Gulf War veterans continue to experience severe symptoms associated with their service, making improved understanding of potential exposure events critically important for their ongoing care.

The methodological frameworks developed through this research will benefit future exposure assessments following chemical incidents or natural disasters. By establishing best practices for integrating multiple data sources in retrospective analysis, the project will improve society's ability to respond to and learn from environmental releases of hazardous materials. These methodological advances will be valuable for emergency response planning, environmental remediation efforts, and long-term health monitoring following chemical incidents.

For the broader scientific community, this research will demonstrate the value of revisiting historical analyses when new data or methodological approaches become available. The project will establish protocols for transparent reassessment that can be applied across scientific disciplines when evaluating the need to reexamine historical conclusions. This model of scientific reassessment has applications far beyond the specific Khamisiyah case.

Educational materials developed from this research will advance scientific literacy regarding chemical agent behavior, atmospheric dispersion, and exposure assessment. These materials will be designed for multiple audiences, including veterans, healthcare providers, environmental scientists, and emergency responders. By translating complex scientific concepts into accessible educational resources, the project will improve public understanding of environmental health risks associated with chemical releases.

The research will provide training opportunities for graduate and undergraduate students in interdisciplinary approaches to complex environmental health questions. Students involved in the project will develop skills in GIS analysis, atmospheric modeling, and scientific communication that will prepare them for careers addressing environmental health challenges. This educational impact extends beyond the immediate project to building future scientific capacity.

By addressing a significant historical controversy with potential implications for veterans' healthcare, this research demonstrates the social responsibility of scientific inquiry. The project exemplifies how rigorous scientific methods can be applied to questions of social justice and institutional accountability, providing a model for science in service of society.

## Research Methods

This study will employ an integrated multi-method approach combining meteorological data analysis, atmospheric dispersion modeling, GIS spatial analysis, and document review. This comprehensive methodology is specifically designed to address the complex interdisciplinary nature of retrospective chemical exposure assessment.

### Meteorological Data Analysis

The foundation of this research is the analysis of newly recovered meteorological data for March 4, 1991. This dataset will be compiled from multiple sources, including:

- The complete meteorological dataset for March 4, 1991, previously claimed nonexistent
- Archived weather records from March 1-15, 1991
- Satellite imagery of regional weather patterns
- Military weather reports and logs from the theater of operations
- Declassified meteorological records from military databases

These diverse data sources will be integrated to develop comprehensive meteorological profiles for both the March 4 and March 10 demolition events. Advanced statistical analysis will be applied to characterize wind patterns, atmospheric stability conditions, temperature profiles, and humidity levels at multiple elevations. The analysis will employ machine learning algorithms specifically designed to identify patterns in complex atmospheric data, particularly focusing on frontal boundary movements and diurnal variations that may have influenced chemical agent dispersion.

This meteorological analysis approach is methodologically superior to previous assessments because it integrates multiple data sources rather than relying primarily on limited upper-air data. The statistical characterization of uncertainty in historical meteorological records will provide a more transparent foundation for subsequent modeling efforts than was possible in earlier analyses.

## Atmospheric Dispersion Modeling

Multiple contemporary atmospheric dispersion models will be employed to develop a more comprehensive understanding of potential chemical agent dispersion patterns than was possible with single-model approaches. The specific models include:

1. HYSPLIT (Hybrid Single-Particle Lagrangian Integrated Trajectory) - for tracking particle trajectories and concentration distributions
2. CALPUFF non-steady-state puff dispersion model - for capturing effects of spatially and temporally varying meteorological conditions
3. AERMOD (American Meteorological Society/Environmental Protection Agency Regulatory Model) - for near-field dispersion assessment

Each model will be configured with:

- The complete meteorological dataset for both demolition events
- Documented release parameters (quantity, composition, release mechanics)
- Physical and chemical properties of sarin and cyclosarin
- Topographical data for the region

The multi-model approach represents a methodological advancement over single-model techniques used in previous assessments. This approach will enable comprehensive uncertainty analysis through comparison of model outputs, providing a more robust understanding of potential dispersion patterns than was previously possible.

## Geographic Information System (GIS) Analysis

GIS technology will be used to integrate spatial data on troop positions, terrain features, and modeled chemical agent dispersion patterns. This analysis will include:

- Digitization of military maps and unit location records from Operation Desert Storm
- Detailed mapping of key locations (Bunker 73, "Pit" area, troop positions)
- Spatial analysis of terrain effects on chemical agent dispersion
- Quantitative assessment of potential troop exposures based on unit positions relative to modeled plume paths

The GIS methodology will incorporate Monte Carlo simulation techniques to address uncertainty in historical positioning data, providing probabilistic exposure assessments rather than binary classifications. This represents a methodological advancement over previous approaches that classified units as either "exposed" or "unexposed" without accounting for positioning uncertainty.

### Document Analysis

Systematic analysis of declassified military records will provide essential context for understanding operational decisions and reported chemical detections. This analysis will include:

- Content analysis of declassified intelligence reports
- Chronological reconstruction of key events and decisions
- Verification of witness accounts against documentary evidence
- Identification of factual discrepancies in official narratives

This documentary analysis will be conducted using established qualitative analysis techniques, with particular attention to source verification and triangulation of information across multiple documents. This approach addresses methodological limitations in previous assessments that may have selectively utilized available documentation.

### Data Management and Ethical Considerations

All data will be cataloged in a secure research database with appropriate metadata documentation. Given the sensitive nature of some declassified materials, special attention will be paid to handling procedures for classified or recently declassified information. No human subjects will be directly involved in this research beyond the analysis of existing declassified documents, but ethical considerations regarding potential implications for veterans' healthcare will be continuously evaluated throughout the project.

This integrated multi-method approach represents the most appropriate methodology for addressing the research questions because it:

1. Incorporates multiple data sources to overcome limitations of any single dataset
2. Employs multiple modeling approaches to develop more robust understanding of potential exposure patterns
3. Integrates quantitative meteorological analysis with qualitative document review to provide comprehensive contextual understanding
4. Explicitly addresses uncertainty through statistical characterization and multi-model comparison

## Project Activities

The research project will be structured around five primary activity areas, each comprising specific tasks designed to systematically address the research questions.

### 1. Data Collection and Organization

This foundational activity involves gathering and organizing all relevant data sources necessary for the analysis:

- **Meteorological Data Compilation**: Systematic collection of the complete March 4, 1991 meteorological dataset from military archives, including surface observations, upper air measurements, and satellite imagery. This will involve coordinating with the National Climatic Data Center and DoD archives to access and digitize historical weather records.
- **Military Document Recovery**: Identification and acquisition of relevant declassified operational records, including unit logs, after-action reports, and chemical detection documentation. This will require working with the National Archives, DoD declassification offices, and FOIA requests to access documents not previously incorporated into analyses.
- **Troop Position Mapping**: Collection and digitization of historical maps and unit location records to establish precise positioning of military units during both demolition events. This will involve working with military historians and archivists to locate and verify troop movement records.
- **Chemical/Physical Data Compilation**: Gathering technical specifications of chemical munitions, including fill quantities, agent composition, and physical properties relevant to dispersion modeling. This will involve consulting with chemical weapons experts and reviewing technical documentation on Iraqi munitions from UNSCOM reports.

Each data source will be cataloged in a comprehensive research database with standardized metadata documentation to ensure transparency and reproducibility. This activity is critical for establishing the factual foundation upon which subsequent analyses will be built.

## 2. Meteorological Analysis and Reconstruction

This activity involves detailed analysis of weather conditions during the study period:

- **Statistical Analysis**: Conducting comprehensive statistical characterization of meteorological conditions, including wind roses, stability classifications, and atmospheric profile reconstruction. Advanced statistical techniques will be applied to characterize temporal and spatial variability in weather conditions.
- **Frontal Boundary Analysis**: Detailed examination of the approaching weather front on March 4, 1991, including its timing, characteristics, and potential effects on chemical agent transport. This will involve synoptic meteorological analysis techniques applied to historical data.
- **Pattern Identification**: Application of machine learning algorithms to identify complex weather patterns affecting the Khamisiyah region during the study period. This novel approach will help identify meteorological phenomena not previously incorporated into dispersion analyses.
- **Comparative Assessment**: Systematic comparison of actual meteorological conditions with assumptions used in previous modeling efforts to identify discrepancies. This will provide a quantitative

basis for understanding how meteorological assumptions affected previous exposure assessments.

This activity will produce comprehensive meteorological profiles for both demolition events that will serve as inputs for subsequent dispersion modeling, addressing Research Question 1.

### 3. Atmospheric Dispersion Modeling

This activity involves developing and comparing multiple dispersion models:

- **Model Configuration**: Setting up each atmospheric dispersion model (HYSPLIT, CALPUFF, AERMOD) with appropriate parameters, including source terms, chemical properties, and meteorological inputs. Model configurations will be documented in detail to ensure reproducibility.
- **Sensitivity Analysis**: Conducting parametric sensitivity analysis to identify key variables affecting dispersion predictions and quantify uncertainty. This will involve systematic variation of input parameters to understand their relative importance.
- **Comparative Modeling**: Running parallel simulations with multiple models to develop a more robust understanding of potential dispersion patterns than single-model approaches can provide. Discrepancies between models will be analyzed to understand sources of uncertainty.
- **Validation Testing**: Evaluating model performance against known reference cases and available field observations to assess reliability. This will include using documented test cases to ensure models are performing as expected.

This activity will produce revised chemical agent dispersion models for both demolition events, addressing Research Question 2 and providing inputs for subsequent geospatial analysis.

### 4. GIS Integration and Exposure Assessment

This activity involves spatial analysis of potential troop exposures:

- **Spatial Database Development**: Creating comprehensive GIS databases integrating digitized military maps, unit locations, terrain data, and modeled plume paths. This will serve as the foundation for all spatial analyses.
- **Exposure Zone Mapping**: Generating time-series maps showing the evolution of potential exposure zones relative to documented troop positions. These will visualize how chemical agent plumes may have interacted with troop positions over time.
- **Probabilistic Assessment**: Developing statistical models to characterize exposure likelihood based on unit positioning and plume characteristics. This approach will move beyond binary exposure classification to provide more nuanced understanding of potential exposures.
- **Comparative Analysis**: Systematically comparing revised exposure estimates with official assessments to identify disparities in potential exposure patterns. This will quantify differences between

revised and official exposure assessments.

This activity will produce comprehensive geospatial analyses of potential troop exposures for both demolition events, addressing Research Question 3.

### 5. Integration and Dissemination

This activity involves synthesizing findings and sharing results:

- **Comprehensive Assessment**: Integrating findings from meteorological analysis, dispersion modeling, and geospatial analysis to develop a comprehensive understanding of the Khamisiyah events. This synthesis will connect findings across methodological approaches.
- **Academic Publication**: Preparing multiple peer-reviewed journal articles detailing methodological approaches and findings. Target journals include the Journal of Exposure Science and Environmental Epidemiology, GeoHealth, and Military Medicine.
- **Educational Materials**: Developing accessible educational resources explaining the findings for veterans, healthcare providers, and the general public. These will include visual explainers, interactive maps, and plain-language summaries.
- **Conference Presentations**: Presenting findings at relevant academic conferences, including the American Meteorological Society Annual Meeting, the American Association of Geographers Conference, and the Military Health System Research Symposium.
- **Policy Brief Development**: Creating targeted policy briefs summarizing implications for veterans' healthcare and research. These will be designed to inform decision-making by relevant authorities.

This activity will ensure that research findings reach both academic audiences and stakeholders who can apply the findings to improve veterans' healthcare.

## Timeline

| Month | Year | Activities |
|---|---|---|
| May | 2025 | • Begin meteorological data compilation<br>• Initiate document recovery process<br>• Establish research database structure |
| June | 2025 | • Complete initial meteorological data compilation<br>• Begin statistical analysis of weather conditions<br>• Continue document collection |
| July | 2025 | • Conduct frontal boundary analysis<br>• Begin digitization of military maps<br>• Complete document collection |
| August | 2025 | • Complete meteorological pattern identification<br>• Begin HYSPLIT model configuration<br>• Complete GIS database structure |
| September | 2025 | • Begin CALPUFF and AERMOD configuration<br>• Complete meteorological statistical analysis<br>• Finalize troop position mapping |
| October | 2025 | • Run initial dispersion models<br>• Begin sensitivity analysis<br>• Start exposure zone mapping for March 4 |
| November | 2025 | • Complete sensitivity analysis<br>• Run comparative modeling simulations<br>• Continue exposure assessment |
| December | 2025 | • Complete dispersion modeling<br>• Finish exposure zone mapping<br>• Begin probabilistic assessment |
| January | 2026 | • Complete probabilistic assessment<br>• Begin comparative analysis with official assessments<br>• Start integration of findings |
| February | 2026 | • Complete comparative analysis<br>• Finalize integration of findings<br>• Begin manuscript preparation |
| March | 2026 | • Complete first manuscript draft<br>• Begin educational material development<br>• Prepare conference abstracts |
| April | 2026 | • Finalize thesis manuscript<br>• Complete educational materials<br>• Prepare policy briefs<br>• Thesis defense |

This timeline allocates appropriate time for each project phase while maintaining flexibility to address challenges that may arise during data collection or analysis. The schedule is realistic and achievable within the one-year timeframe, with equal emphasis on thorough analysis and effective dissemination of findings.

## Data Management Plan

### Data Types and Collection

This research will utilize multiple types of data, including:

1. **Meteorological Data**: Historical weather observations, including surface measurements, upper air soundings, and satellite imagery. These will be obtained from National Climatic Data Center archives,

military weather records, and declassified meteorological logs.

2. **Military Operational Records**: Declassified unit logs, after-action reports, and operational maps related to the Khamisiyah demolitions. These will be obtained through the National Archives, FOIA requests, and DoD declassification offices.
3. **Geospatial Data**: Digitized maps, unit location coordinates, and terrain data for the Khamisiyah region. These will be derived from military maps, operational records, and satellite imagery.
4. **Model Output Data**: Results from atmospheric dispersion simulations, including concentration fields, particle trajectories, and exposure zones. These will be generated through the research process.

Data collection will focus on ensuring completeness and documentation of provenance. No human subjects data will be collected for this research.

## Data and Metadata Standards

The project will employ the following standards:

1. **Meteorological Data**: Data will be formatted according to World Meteorological Organization (WMO) standards, using standardized units and formats for observations. Metadata will include observation time, location, instrument type, and measurement uncertainty.
2. **Geospatial Data**: All geospatial data will adhere to Federal Geographic Data Committee (FGDC) standards for metadata, including coordinate systems, datum information, and accuracy assessments. Data will be stored in standard GIS formats (shapefile, GeoTIFF) for interoperability.
3. **Model Input/Output**: Model parameters and outputs will be documented following the Air Quality Model Evaluation International Initiative (AQMEII) protocols, with standardized documentation of input parameters, model versions, and output metrics.
4. **Documentation Records**: Archival records will be cataloged using Dublin Core metadata standards, including source information, acquisition date, and declassification status.

These standards will ensure that data is well-documented, interoperable, and accompanied by appropriate contextual information.

## Data Storage and Protection During the Project

During the active research phase, data will be stored and protected as follows:

1. **Primary Storage**: All research data will be stored on the University of Missouri's secure research storage infrastructure, which provides redundant backup, encryption, and access controls.
2. **Working Copies**: Working data copies will be maintained on password-protected research computers with regular synchronization to the primary storage.

3. **Version Control**: All analysis scripts, model configurations, and documentation will be maintained under version control using Git, with regular commits to track changes and maintain data provenance.
4. **Access Controls**: Access to research data will be limited to the principal investigator and authorized research team members. Access logs will be maintained for security and provenance tracking.
5. **Data Classification**: Data will be classified according to sensitivity, with appropriate security measures applied to declassified military records that may contain sensitive information.

These measures will ensure data security while allowing efficient research workflows.

## Data Ownership and Long-term Management

Data ownership and management responsibilities are defined as follows:

1. **Primary Data Ownership**: The University of Missouri will retain intellectual property rights for derived data products, including model outputs, geospatial analyses, and research findings.
2. **Archival Document Rights**: Original government documents will remain the property of their originating agencies, with research use governed by applicable regulations for declassified materials.
3. **Long-term Repository**: Upon project completion, research data will be deposited in the following repositories:
   - Meteorological and model data: NOAA's National Centers for Environmental Information (NCEI)
   - Geospatial data: OpenGeoLab repository at the University of Missouri
   - Document catalogs and finding aids: Gulf War Veterans Resource Collection
4. **Data Stewardship**: The Department of Geography at the University of Missouri will maintain stewardship of the complete research dataset for a minimum of ten years following project completion, ensuring continued availability for verification and follow-up research.

## Data Access and Reuse

The following data access policies will be implemented:

1. **Open Access Commitment**: All derived research data, analysis scripts, and model configurations will be made publicly available following thesis completion, subject to University of Missouri data policies.
2. **Access Mechanisms**: Data will be accessible through:
   - Repository direct download
   - API access where supported by repositories
   - Direct request to the principal investigator
3. **Licensing**: Dataset will be released under a Creative Commons Attribution 4.0 International License (CC BY 4.0), allowing broad reuse with appropriate attribution.

4. **Usage Restrictions**: Usage limitations will apply only to sensitive declassified materials, which will be governed by original declassification terms. For these materials, catalog information will be provided with instructions for accessing the original documents through appropriate government channels.
5. **Embargo Period**: A brief embargo period of six months following thesis completion will allow for preparation of primary research publications before data release.

These policies balance open scientific access with appropriate protections for sensitive information, maximizing the research's long-term value while respecting security considerations.

## Letters of Collaboration and Permits

[Note: The following institutions have provided letters of support for this research project. These letters are attached as appendices to the proposal.]

1. **University of Missouri Department of Geography** - Letter confirming institutional support, data storage resources, and GIS laboratory access for the duration of the project.
2. **National Archives and Records Administration** - Letter acknowledging the research purpose and confirming cooperation with document retrieval requests related to declassified Gulf War records.
3. **Gulf War Veterans Resource Center** - Letter of collaboration offering access to their document collections and assistance with disseminating findings to the veteran community.
4. **Air Resources Laboratory, National Oceanic and Atmospheric Administration** - Letter confirming technical support for HYSPLIT modeling applications used in the research.
5. **University of Missouri Institutional Review Board** - Letter confirming that the proposed research, focusing exclusively on document analysis and modeling without human subjects research, is exempt from IRB review requirements.

[Note: No permits are required for this research as it involves analysis of existing data rather than field collection activities or human subjects research.]